

Test Results Document

Horizon AI — Multi-Agent Java Refactoring Pipeline

268 / 268 tests passing — 100% pass rate

1. Executive Summary

Suite	Tests	Passed	Failed	Pass Rate
Backend Unit	167	167	0	100%
Backend Integration	12	12	0	100%
Frontend Unit	85	85	0	100%
Frontend Integration	4	4	0	100%
Total	268	268	0	100%

Test execution time: Backend ~2.5s, Frontend ~4.0s
Total test files: 31 (18 backend + 13 frontend)
Lint status: 0 errors (ruff, eslint). TypeScript: 0 errors (tsc).

2. Backend Unit Test Results

2.1 Summary by Module

Module	Tests	Passed
OrchestrationConfig	6	6
Types & Schemas	8	8
ResponseParser	20	20
ASTMatcher	12	12
Validator	25	25
AgentService	10	10
ClientConnection	8	8
DatabaseManager	12	12
MessageRouter	8	8
Orchestrator	7	7
Phase 2 — Strategy	12	12
Phase 3 — Execution	10	10

Phase 4 — Validation	6	6
Phase 5 — Adjudication	5	5
Phase 6 — Finalization	5	5
Formatters	6	6
Code Utils	7	7

All backend unit tests use the AAA (Arrange-Act-Assert) pattern.

Pure functions: ResponseParser, ASTMatcher, Formatters, Code Utils — no mocking.

Mocked services: AgentService, Router, Connection, Orchestrator — AsyncMock/MagicMock for external deps.

Domain logic: Validator, Phase 2-6 — real javalang/lizard libraries.

3. Backend Integration Test Results

Test ID	Scenario	Result
TC-IT-001	GET /health returns 200 with timestamp	PASS
TC-IT-002	GET /api/history returns empty list	PASS
TC-IT-004	GET /api/history/:id returns 404 for missing	PASS
TC-IT-006	PATCH /api/history/:id rejects empty title	PASS
TC-IT-008	DELETE /api/history/:id returns 404 for missing	PASS
TC-IT-009	WebSocket /ws accepts connection	PASS
TC-IT-011	WebSocket halt notification via router	PASS
TC-IT-012	WebSocket malformed JSON rejection	PASS
TC-PL-SUCCESS	Full pipeline completes, session persisted	PASS
TC-PL-RENAME	Rename + delete session after pipeline	PASS
TC-PL-HALT	User halt during generation	PASS
TC-PL-ABORT	Strategy retry circuit breaker	PASS

FastAPI TestClient with in-memory SQLite. Real Validator, real DatabaseManager, real 6-phase pipeline. Only AgentService.generate is mocked.

4. Frontend Test Results

4.1 Summary by Module

Module	Tests	Passed
lib/utils (cn)	6	6
lib/constants	6	6
lib/parseStatusInfo	12	12
lib/formatStatusContent	8	8
lib/javaFormatter	9	9
lib/indentation	7	7
lib/buildMetrics	6	6
lib/schemas (Zod)	8	8
store/useChatStore	15	15
hooks/useOrchestrationSocket	3	3
ErrorBoundary	4	4
Terminal	3	3
ChatWorkspace	2	2

Logic tests (utils, parsers, formatters): pure functions in isolation.

Store tests: real Zustand state with mocked fetch().

Hook tests: renderHook with mocked WebSocket constructor.

Component tests: React Testing Library with mocked stores.

5. CI Pipeline Results

Job	Tool	Result
Lint	ruff	All checks passed
Type check	mypy	Passed
Unit tests	pytest tests/unit/	167/167 passing
Integration tests	pytest tests/integration/	12/12 passing
Frontend lint	eslint	0 errors, 0 warnings
Frontend type check	npx tsc --noEmit	0 errors
Frontend tests	vitest run	89/89 passing
Frontend build	next build	Build complete